

Topology of the real line – 3

Compactness and covering arguments

Homework

8 сентября 2026 года

Theoretical minimum. Open covers and compactness; the four equivalent forms of the Heine–Borel theorem; the Lebesgue number lemma; the Lindelöf property; the finite intersection property; the rising sun lemma and the finite Vitali covering lemma as the two standard ways to turn a pointwise condition into a uniform one.

1. Prove the finite Vitali covering lemma: from any finite family of intervals one can select a pairwise disjoint subfamily whose total length is at least one third of the measure of the union. Show that the constant $\frac{1}{3}$ cannot be replaced by anything larger than $\frac{1}{2}$.
2. (MIPT) Two players play on a set $X \subseteq \mathbb{R}$, making countably many moves. At the n -th move the first player names a length $d_n > 0$ and the second player places a segment of length d_n on the line. The second player wins if the segments cover X . For which sets X does the second player have a winning strategy?
3. (MIPT) Let K be a compact subset of Euclidean space and let $f: K \rightarrow K$ satisfy $|f(x) - f(y)| \geq |x - y|$ for all $x, y \in K$. Prove that every point of K is a limit point of its own orbit $\{f^n(x)\}$, and deduce that f preserves distances.
4. Prove that a decreasing sequence of non-empty compact connected subsets of $\mathbb{R}^n$ has connected intersection, and construct a decreasing sequence of closed connected subsets of the plane whose intersection is non-empty but disconnected. Note that non-emptiness of the intersection is not what is at stake here: it is connectedness that needs compactness.

5. Prove Cousin's lemma: for every function δ assigning a positive number to each point of $[a, b]$ there is a tagged partition $a = x_0 < \cdots < x_n = b$ with tags t_i satisfying $[x_{i-1}, x_i] \subseteq (t_i - \delta(t_i), t_i + \delta(t_i))$. Then derive the Heine–Borel theorem from it.
6. Prove that if A is compact, B is closed and $A \cap B = \emptyset$, then $\text{dist}(A, B) > 0$, and give two disjoint closed sets in $\mathbb{R}$ at distance zero.
7. Prove that every compact $K \subseteq \mathbb{R}$ is the intersection of a decreasing sequence of finite unions of closed intervals, and deduce that for every $\varepsilon > 0$ the set K has a finite ε -net.
8. Prove the finite intersection property characterisation of compactness, and use it to show that a family of compact sets in which every finite subfamily has a common point has a common point, while the same fails for closed unbounded sets.
9. Prove that from every open cover of $[a, b]$ by intervals one can extract a finite subcover in which no point of $[a, b]$ belongs to more than two of the chosen intervals.
10. Prove that $K \subseteq \mathbb{R}$ is compact if and only if every continuous function on K is bounded, and also if and only if every continuous function on K attains its supremum.